

SPOT+ FAMILY

REST API USER GUIDE

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📌 Advanced Digital Pyrometer Series



LAND
AMETEK®



QUALITY CUSTOMER SOLUTIONS

Health and Safety Information



Read all of the instructions in this booklet - including all the **WARNINGS** and **CAUTIONS** - **before** using this product. If there is any instruction which you do not understand, **DO NOT USE THE PRODUCT**.

Safety Signs



WARNING

Indicates a potentially hazardous situation which, if not avoided, could result in death or personal injury.



CAUTION

Indicates a potentially hazardous situation which, if not avoided, could result in minor or moderate injury to the user or users, or result in damage to the product or to property.

NOTE

Indicates a potentially hazardous situation which, if not avoided, could result in damage or loss of data.

Signs and Symbols used on equipment and Documentation



Caution, risk of electric shock.



Caution, attention to possibility of risk of damage to the product, process or surroundings. Refer to instruction manual.



Caution, hot surface.



Protective Conductor Terminal.



Observe precautions for handling electrostatic discharge sensitive devices.

Equipment Operation

Use of this instrument in a manner not specified by AMETEK Land may be hazardous. Read **and understand** the user documentation supplied **before** installing and operating the equipment.

The safety of any system incorporating this equipment is the responsibility of the assembler.

Protective Clothing, Face and Eye Protection

It is possible that this equipment is to be installed on, or near to, machinery or equipment operating at high temperatures and high pressures. Suitable protective clothing, along with face and eye protection must be worn. Refer to the health and safety guidelines for the machinery/equipment before installing this product. If in doubt, contact AMETEK Land.



Wear Protective Gloves



Wear Protective Clothing



Wear Eye Protection



Wear Ear Protection



Wear Safety Boots



Wear Face Protection

Electrical Power Supply

Before working on the electrical connections, all of the electrical power lines to the equipment must be isolated. All the electrical cables and signal cables must be connected exactly as indicated in these operating instructions. If in doubt, contact AMETEK Land.

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Storage

The instrument should be stored in its packaging, in a dry sheltered area.

The maximum storage temperature is 10°C (18°F) higher than the maximum operating temperature.

The minimum storage temperature is 10°C (18°F) lower than the minimum operating temperature.

Refer to the Technical Specification for details of the operating temperature limits.

Unpacking

Check all packages for external signs of damage. Check the contents against the packing note.

Lifting Instructions

Where items are too heavy to be lifted manually, use suitably rated lifting equipment. Refer to the Technical Specification for weights. All lifting should be carried out in accordance with local and national regulations.

Return of Damaged Goods

IMPORTANT If any item has been damaged in transit, this should be reported to the carrier and to the supplier immediately. Damage caused in transit is the responsibility of the carrier not the supplier.

DO NOT RETURN a damaged instrument to the sender as the carrier will not then consider a claim. Save the packing with the damaged article for inspection by the carrier.

Return of Goods for Repair

If you need to return goods for repair please contact our Customer Service Department for details of the correct returns procedure.

Any item returned to AMETEK Land should be adequately packaged to prevent damage during transit.

You must include a written report of the problem together with your own name and contact information, address, telephone number, email address etc.

Design and Manufacturing Standards

The Quality Management System of Land Instruments International is approved to BS EN ISO 9001 for the design, manufacture and on-site servicing of combustion, environmental monitoring and non-contact temperature measuring instrumentation.

Registered ISO9001 Management System approvals apply in the USA.

UK Calibration Laboratory: UKAS 0034.

USA Calibration Laboratory: ANAB Accredited ISO/IEC 17025.

National Accreditation Board for Testing and Calibration Laboratories approvals apply in India.

Operation of radio transmitters, telephones or other electrical/electronic devices in close proximity to the equipment while the enclosure doors of the instrument or its peripherals are open, may cause interference and possible failure where the radiated emissions exceed the EMC directive.

The protection provided by this product may be invalidated if alterations or additions are made to the structural, electrical, mechanical, pneumatic, software or firmware components of this system. Such changes may also invalidate the standard terms of warranty.

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1

INTRODUCTION

1.1 Introduction

This User Guide explains how to communicate with a SPOT+ pyrometer using the REST API.

AMETEK Land's SPOT+ range is a family of fully-featured smart pyrometers for non-contact temperature measurements. Available in a number of temperature ranges and models, SPOT+ pyrometers provide immediate local display and onboard analogue I/O together with multiple communication options over Ethernet: Embedded Webserver, Modbus TCP/IP, REST API (http) and EthernetIP.

The REST (REpresentational State Transfer) API is a protocol for ethernet communication via http GET and POST or PUT requests. As it is the protocol used by every website on the internet, it is extremely easy and flexible to use for software development.

The SPOT+ REST API is designed to allow read and write of only those parameters required during normal operation. The SPOT+ embedded webserver should be used for initial instrument configuration – setting parameters such as the IP address or process window correction that will remain fixed after initial commissioning. Information on configuration of the SPOT+ using the embedded webserver is given in the SPOT+ User Guide.

REST API commands take the form of an http PUT or GET command directed to a URL on the SPOT+ thermometer webserver. The SPOT+ returns a JSON file or a single parameter, depending on the request.

	Request	Example Reply
Read multiple values	GET http://ipaddress/node	Formatted JSON containing all values. Eg:={ "item1":1234, "item2":56, "item3":789 }
Read single parameter	GET http://ipaddress/node?p=item1	1234
Write single parameter	PUT http://ipaddress/node?p=item1 BODY=0.567	0.567

The SPOT API has three JSON nodes: output, buffer, control. Images can also be read directly from the SPOT+, already parsed as jpg files. Available parameters are listed in Section 2 and their available commands given in 3 of this document.

Before REST applications can change any parameters on the instrument during normal operation, the ethernet security level must be configured. Ethernet security settings for the SPOT+ are described in section 4.

Example Javascript code is given in the final section of this document. A sample webpage using the REST API is available to download and copy from the AMETEK Land website, www.ametek-land.com/spot.

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INDEX OF SPOT+ REST API PARAMETERS

2.1 SPOT+ Key Output Values

The SPOT+ key output values can be read individually by a GET request to `http://[ipaddress]/output?p=[parameter]` or collectively as a GET request to `http://[ipaddress]/output`

Parameter	Read	Write	Definition and Range
temperature	✓	✗	Target temperature output, after time functions and instrument mode calculations have been applied. Values 0-6500 in degrees C or F (depending on instrument configuration), reported to 1 decimal place. Over and under-range values are transmitted as 6553.5 and 6553.4 respectively.
itemperature	✓	✗	Internal temperature of the instrument. Values from 0 to 100C or 212F (depending on instrument configuration) reported to 1 decimal place.
alarmstatus	✓	✗	Alarm status byte. Any value >0 causes the CMD Out relay connections to the SPOT+ to be closed if normally open (or vice versa, depending on instrument configuration). Individual bits are set to 1 if the alarm is configured and the alarm threshold is exceeded: Bit 0: low ambient temperature Bit 1: high ambient temperature Bit 2: low target temperature Bit 3: high target temperature Bit 4: low signal% *
d1temperature *	✓	✗	Measured temperature from the Mono1 channel of a Ratio or Application SPOT+. Values 0-6500 in degrees C or F (depending on instrument configuration), reported to 1 decimal place.
d2temperature *	✓	✗	Measured temperature from the Mono1 channel of a Ratio or Application SPOT+. Values 0-6500 in degrees C or F (depending on instrument configuration), reported to 1 decimal place.

Parameter	Read	Write	Definition and Range
signalpc *	✓	✗	Infrared signal strength as a percentage of that expected from a blackbody target of the same temperature as the target. Reduced signal strength can be due to target emissivity less than 1 or obscuration of the sight path. Values integer % values from 0-100 %.
e1out **	✓	✗	Target emissivity for the shorter wavelength channel calculated by an Application pyrometer. Values from 0 to 1.2 reported to 3 decimal places.
e2out **	✓	✗	Target emissivity for the longer wavelength channel calculated by an Application pyrometer. Values from 0 to 1.2 reported to 3 decimal places.

* Available on Ratio or Application specific models

** Available on Application specific models.

2.2 SPOT+ Fast Temperature Output Buffer

The SPOT+ temperature output buffer can be read together with its pointer as a GET request to [http://\[ipaddress\]/buffer](http://[ipaddress]/buffer)

Parameter	Read	Write	Definition and Range
buffer	✓	✗	Rolling 100 item buffer of temperature measurements, for use with fast response times. Every output interval, the latest temperature measurement is copied to the next position in the buffer, and this position is saved as the buffer pointer. Values 0-6500 in degrees C or F (depending on instrument configuration), reported to 1 decimal place.
pointer	✓	✗	Position in the buffer of the most recent measurement. Integer values 0-99.

2.3 SPOT+ Camera Images

The latest jpg image from the SPOT+ camera can be read as a GET request to [http://\[ipaddress\]/image.jpg](http://[ipaddress]/image.jpg)

Parameter	Read	Write	Definition and Range
image	✓	✗	Most recent image from the SPOT+ visible light camera, in jpg format.

2.4 SPOT+ Control Parameters

The SPOT+ control parameters can be read and written individually as GET and PUT requests to `http://[ipaddress]/control/p=[parameter]`.

Parameter	Read	Write	Definition and Range
emissivity1	✓	✓	User-entered emissivity value to correct measurements of a Mono pyrometer, or the Mono1 channel of a Ratio or Application Pyrometer. Values from 0.05 to 1.2 in steps of 0.001
emissivity2 *	✓	✓	User-entered emissivity value to correct measurements of the Mono2 channel of a Ratio of Application Pyrometer. Values from 0.05 to 1.2 in steps of 0.001
Bgdtemperature	✓	✓	User-entered value of background temperature to correct target temperature measurements. Integer values of 0 to 6500 in degrees C or F (depending on instrument configuration). Note that background temperature correction is not available in Ratio mode.
focus	✓	✓	Focus distance from the front of the instrument to the target. Integer values from 300 to 10000 in mm.
led	✓	✓	Command to start or stop the LED flashing. Writing a 1 will start the LED flashing. Writing a 0 will stop it flashing, if it was flashing. Alternatively the LED will timeout 30s after the last instruction to start flashing.
cmdin	✓	✓	Trigger command that can be used to reset the Peak Picker or initiate Sampling, depending on instrument configuration. Writing a 1 will activate and writing a 0 will clear the digital command input.
errorcode	✓	✗	Hexadecimal number ??
info	✓	✗	A string detailing instrument type, mode and current settings, as displayed on the info page of the instrument and webserver

Parameter	Read	Write	Definition and Range
appnumber **	✓	✓	The index of the current App Mode. Integer values from 1 up to the number of Apps that are installed on the instrument.
appoffset **	✓	✓	The calibration offset to the current App. Integer values from -2000 to 2000.
reftemperature	✗	✓	Reference temperature used to adjust the emissivity of a Mono or Ratio pyrometer, or tune the current App of an Application pyrometer by setting a calibration offset value. Integer temperature values from 0 to 6500 degrees in C or F, depending on instrument configuration.

* Available on Ratio or Application specific models

** Available on Application specific models.

3

SPOT+ COMMAND REFERENCE

3.1 SPOT+ Key Output Values

The principal measurement outputs from the SPOT+ thermometer can be read individually or altogether in a single 'read multiple' request. When a 'read multiple' request is used, the SPOT+ returns a formatted JSON file containing all parameters relevant to the particular instrument. Example communications are presented here for each type of instrument.

3.1.1 Read Multiple: Mono Instruments

Mono instruments use only one wavelength channel so only return target temperature, internal instrument temperature and alarm status.

Request	Example Reply
GET http://[ipaddress]/output	{ "temperature":512.1, "itemperature":41.2, "alarmstatus":0 }

3.1.2 Read Multiple: Ratio Instruments

Ratio instruments use two wavelength channels and additionally return individual channel measurements and the measured infrared signal strength.

Request	Example Reply
GET http://[ipaddress]/output	{ "temperature":512.1, "itemperature":41.2, "alarmstatus":0, "d1temperature":400.0, "d2temperature":350.5, "signalpc":10, }

3.1.3 Read Multiple: Application Instruments

Application specific instruments additionally measure target emissivity at both measurement wavelengths.

Request	Example Reply
GET http://[ipaddress]/output	{ "temperature":512.1, "itemperature":41.2, "alarmstatus":0, "d1temperature":400.0, "d2temperature":350.5, "signalpc":10, "e1out":0.101, "e2out":0.975 }

3.1.4 Temperature Measurements

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/output?p=temperature	512.1

3.1.5 Internal Temperature

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/output?p=itemperature	41.2

3.1.6 Alarm Status

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/output?p=alarmstatus	0

3.1.7 Individual Detector Readings (Ratio and Application Pyrometers only)

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/output?p=d1temperature	400.0

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/output?p=d2temperature	350.5

3.1.8 Signal % (Ratio and Application Pyrometers only)

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/output?p=signalpc	10

3.1.9 Emissivity Measurements (Application Pyrometers only)

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/output?p=e1out	0.101

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/output?p=e2out	0.975

3.2 SPOT+ Fast Temperature Buffer

If the SPOT+ output time is set to <50ms, it is advisable to read an array of the most recent temperature measurements from the internal measurement buffer in batches at intervals of 50ms or more, to reduce network traffic and avoid missing data. The SPOT+ temperature buffer structure is explained in Table 2.

The buffer pointer is required at the same time as the array of recent temperature measurements from the buffer, so a 'read multiple' request must be used.

In this example, the SPOT+ response time is set to 1ms, and it is looking at a slowly cooling target. The latest measurement is the '510.8' value in the fourth position in the array.

NOTE - *Before firmware version 40.33 Buffer matched the MODBUS output so is scaled up by 10 with no decimal place for the temperature reading. Firmware version 40.33 and later responds with the temperature scaled correctly.

Request	Example Reply
GET http://[ipaddress]/buffer	{ "buffer": [512.1, 511.7, 511.2, 510.8, 545.2, 545.1, ...] "pointer": 3 }

3.3 SPOT+ Camera Images

Camera images can be read directly from the SPOT+ with a GET request.

Request	Example Reply
GET http://[ipaddress]/image	[camera image in standard jpg format]

3.4 SPOT+ Fast Temperature Buffer

SPOT+ control parameters can only be read and written individually. The 'lock' function on the SPOT+ webserver can be used to prevent unauthorized writing of SPOT+ control parameters. When the SPOT+ webserver is locked, it will be possible to read but not write to the control parameters. [To unlock the webserver, enter password '230' on the Advanced settings page.]

3.4.1 Emissivity

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/control?p=emissivity1	1.000
Write single value	PUT http://[ipaddress]/control?p=emissivity1 BODY=0.76	0.760

The emissivity of the Mono2 channel can also be configured for Ratio Pyrometers.

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/control?p=emissivity1	1.000
Write single value	PUT http://[ipaddress]/control?p=emissivity1 BODY=0.76	0.760

3.4.2 Background Temperature

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/control?p=bgdtemperature	905
Write single value	PUT http://[ipaddress]/control?p=bgdtemperature BODY=890	890

3.4.3 Focus Position

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/control?p=focus	1000
Write single value	PUT http://[ipaddress]/control?p=focus BODY=500	500

3.4.4 LED

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/control?p=led	0
Write single value	PUT http://[ipaddress]/control?p=led BODY=1	1

3.4.5 Command Input

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/control?p=cmdin	0
Write single value	PUT http://[ipaddress]/control?p=cmdin BODY=1	1

3.4.6 Error Code

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/control?p=errorcode	0x0000

3.4.7 Information

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/control?p=info	SPOT+ R160 Ratio PP

3.4.8 App Number (Application Pyrometers Only)

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/control?p=appnumber	2
Write single value	PUT http://[ipaddress]/control?p=appnumber BODY=1	1

3.4.9 App Calibration Offset (Application Pyrometers Only)

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/control?p=appoffset	-1120
Write single value	PUT http://[ipaddress]/control?p=appoffset BODY=500	500

3.4.10 Reference Temperature

Mode	Request	Example Reply
Read single value	GET http://[ipaddress]/control?p=info	SPOT+ R160 Ratio PP

3.5 Error Responses

If the SPOT+ cannot process the instruction, the appropriate http error code will be returned in response to the GET or PUT response. The response of the message will give additional information.

http error code	Message
400 Bad Request	[parameter] not recognised
401 Unauthorized	Unauthorized access
403 Forbidden	[value] out of range
404 Not Found	[node] not recognised

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ETHERNET SECURITY SETTINGS

There are two security modes for SPOT+ Webserver and REST API communication: Secure or Open.

'Secure' is the recommended setting for SPOT+ control over ethernet in secure applications. Several security features are enabled:

- A handshake is required when a control application first tries to write a parameter. The REST application must know the SPOT+ password.
- The password is hashed and used in authentication, but not transmitted in its entirety except during password change via the webserver. The hashed authentication code changes with every write to prevent packet copying attacks.
- While one application is actively controlling a SPOT+, it allows reads but not writes from a second source.
- Writes are limited to 1 per second to prevent password blitzing or denial of service attacks.
- Incorrect password attempts will be counted.

SPOT+ pyrometers are shipped in the 'Secure' state. The security mode can be switched via the back panel of the display, or via the webserver. In Secure Mode, a password is required to change settings on the SPOT+. The default password is the serial number of the instrument, but this should be changed on first use.

The operational differences between the modes are described below.

4.1 Open Communications

When the user changes to open, the SPOT+ will behave like the earlier SPOT models, freely accepting instructions from any source.

No padlock symbol will be shown on the webserver.

4.2 Secure Communications

Unless the password is entered, the SPOT+ is effectively 'Read Only'. All requested data will be sent but parameters cannot be changed over ethernet, even using the webserver. REST applications will respond to GET requests as expected, but PUT requests will return a '401 Unauthorized' error.

Before password entry, the yellow padlock symbol will be shown on each webserver page. Clicking on the padlock symbol will route the user to the Security page of the webserver where the password can be entered to unlock the device. Ethernet communications can then be set to 'Secure' or 'Open' to permit writes from REST API applications.

Once the password has been entered a green open padlock symbol will be shown on each webserver page. Settings can be changed on the device.

4.2.1 REST API 'Handshake' for Secure Mode Communication

Communications in secure mode use a simplified form of digest authentication.

Both SPOT+ and the client REST application know and store the password, but it is not directly transmitted. The password is hashed with a 'nonce' value (number used once) generated by the SPOT+ during the 'handshake' on first write request, and a nonce counter managed by the client application and incremented with each request.

The client hashes password, nonce and counter together before each PUT request, and sends the nonce, counter and hash in the header of that request.

The SPOT+ performs its own hash using the nonce and counter received and checks against the client hash value. The PUT request is only processed if the nonce is recognized as an established connection, and the hashes match.

On first connection, a 'handshake' is required. The client's first PUT request will typically be using nonce and counter values of 0. Hashed values would match but the nonce will not be recognized. The SPOT+ will generate a new nonce and reply to the PUT request with a '401 Unauthorized' error, with the new nonce in the header. The client must update its nonce value, and hash, and retransmit the PUT request with this new nonce, counter and hash value.

The client can use the same nonce, hashed with an incremented counter, for subsequent PUT requests, though nonce values are time limited and lost on power cycling, so applications must repeat the handshake periodically.

A second client application that knows the password can connect in the same way and will be given its own nonce value. The SPOT+ can support writes from multiple active REST connections.

GET requests do not need any authentication. ⇐⇐⇐

5

EXAMPLE JAVASCRIPT CODE

The code in this section is taken from the SPOT+ REST API Example Webpage, available to download from www.ametek-land.com/spot

5.1 Read All Outputs

```
function readOutputs(){
    //create a new XMLHttpRequest to read temperature measurements
    var outhttp = new XMLHttpRequest();
    outhttp.open("GET", "http://10.1.10.50/output", true);
    //configure the header to request JSON data
    outhttp.setRequestHeader("Content-type", "application/json;charset=utf-8");

    //write a function to handle the reply
    outhttp.onreadystatechange = function() {
        if (this.readyState == 4 && this.status == 200) {
            window.alert(this.responseText);
        }
    };

    //send the request
    outhttp.send();
}
```

5.2 Read a Single Output

```
function readTemperature(){
    //create a new XMLHttpRequest to read temperature measurements
    var temphttp = new XMLHttpRequest();
    temphttp.open("GET", "http://10.1.10.50/output?p=temperature", true);
    //configure the header to request JSON data
    temphttp.setRequestHeader("Content-type", "application/json;charset=utf-8");

    //write a function to handle the reply
    temphttp.onreadystatechange = function() {
        if (this.readyState == 4 && this.status == 200) {
            //if this is the formal reply, interpret it as an integer of 10x C
        }
    }
}
```

```

        //format temperature to 1 decimal place or `----.-` for an over- or
under-range value.
        let t = parseInt(this.responseText);
        //overrange values are transmitted as 65535 and under-range values as
65534
        if (t>6553) t="----.-";
        //display the latest value in the webpage
        document.getElementById("temperature").innerHTML = t + "C";
        //call this function again in half a second's time.
        this.onreadystatechange=null;
        setTimeout(readTemperature,500);
    }
};

//send the request
temphttp.send();
}

```

5.3 Read the Image Stream

```

function readImage(){
    //create a new XMLHttpRequest to read the next image
    var imagehttp = new XMLHttpRequest();
    imagehttp.open("GET", "http://"+ipaddress+"/image.jpg", true);
    //configure the response type to handle images
    imagehttp.responseType = 'blob';

    //write a function to handle the reply
    imagehttp.onreadystatechange = function() {
        if (this.readyState == 4 && this.status == 200) {
            //if this is the formal reply, display the image in the webpage
            let jpg = URL.createObjectURL(this.response);
            document.getElementById("image").src = jpg;
            //close the URL to release the memory allocation
            URL.revokeObjectURL(jpg);
            //The function calls itself on completion, to get the next image
            readImage();
        }
    };

    //send the request
    imagehttp.send();
}

```

5.4 Read the Temperature Buffer

```
function readBuffer(){
  //create a new XMLHttpRequest to read the temperature buffer
  var bufferhttp = new XMLHttpRequest();
  bufferhttp.open("GET", "http://10.1.10.50/buffer", true);
  //configure the header to request JSON data
  bufferhttp.setRequestHeader("Content-type", "application/json;charset=utf-8");

  //write a function to handle the reply
  bufferhttp.onreadystatechange = function() {
    if (this.readyState == 4 && this.status == 200) {
      //update the graph to display the latest buffer
      let reply = JSON.parse(this.responseText.replace(/(\r\n|\n|\r)/g,""));
      let pointer = Number(reply.pointer);
      let points = "";
      let min = Math.min(...reply.buffer); //scale the graph from min to max of the
buffer
      let range = Math.max(...reply.buffer) - min;
      for (var i=0;i<100;i++){ //loop through 100 item buffer
        let j = pointer+i+1; //start at the oldest, the next to be over-written
        if (j>99) j-=100; //then from the beginning of the circular buffer
        let x = i*5+2; //scale the 100 item buffer within a 500 pixel wide svg
        let y = 100-parseInt((reply.buffer[j]-min)/range*100); //scale min to max
100 pixels high
        points += x + "," + y + " ";
      }
      document.getElementById("graph").setAttribute("points", points);
      //read at a faster rate for 1ms SPOT+ response times as ethernet messages
can be delayed
      setTimeout(readBuffer,100);
    }
  };
}

//send the request
bufferhttp.send();
}
```

5.5 Write to a Control Input

The protocol for writing parameters over http depends on the Security Mode of the SPOT+. The following code demonstrates how to switch the LED on and off in Open and in Secure mode.

5.5.1 Open Ethernet Communications

```
function setLED(){
  //this function writes a new value to the LED state
  //open a new XMLHttpRequest
  var ledhttpwrite = new XMLHttpRequest();
  ledhttpwrite.open("PUT", "http://10.1.10.50/control?p=led", true);

  //create a function to handle replies
  //this is only called if the authentication fails, to initiate a new 'handshake'
  ledhttpwrite.onreadystatechange = function() {
    if (this.readyState == 4 && this.status == 200) {
      if (this.responseText.includes("OK")) {
        //the request is complete
      }
    }
  }

  //send the instruction to set the new LED state
  ledhttpwrite.send(wantLED);
}
```

5.5.2 Secure Ethernet Communications

In Secure Mode, the REST application must store additional variables for the password and nonceCounter. The SPOT+ generates a nonce the first time the write is attempted. Each write request must be authenticated with headers containing an encoded JSON web token built from a nonce, nonceCounter and password.

A Javascript file containing functions to perform the encoding are available to download from the [ametek-land.com](http://www.ametek-land.com) website alongside the example SPOT+ REST API webpage. This file, 'SecureModeFunctions.js' is referenced at the top of the REST API page.

```
<script type="text/javascript" src="SecureModeFunctions.js"></script>
function setLED(){
  //this function writes a new value to the LED state
  //open a new XMLHttpRequest
  var ledhttpwrite = new XMLHttpRequest();
  ledhttpwrite.open("PUT", "http://10.1.10.50/control?p=led", true);
```

```

//configure the header to request JSON data
ledhttpwrite.setRequestHeader("Content-type", "application/json;charset=utf-8");

//read the password
let password = document.getElementById("pass").value;
//create a JSON Web Token from the password, nonce and incremented nonce
counter
let jwt_bytes = createJWT(password, nonce, ++nonceCounter);
let signedSecret = base64FromArrayBuffer(jwt_bytes);
signedSecret = signedSecret.replaceAll("=", "");
let jwt = jwt_header + "." + jwt_payload + "." + signedSecret;
//request headers are the full JSON web token, the nonce and nonce counter
ledhttpwrite.setRequestHeader("Authorization", "Bearer " + jwt);
ledhttpwrite.setRequestHeader("nonce", nonce);
ledhttpwrite.setRequestHeader("nc", nonceCounter);

//create a function to handle replies
//this is only called if the authentication fails, to initiate a new 'handshake'
ledhttpwrite.onreadystatechange = function() {
if (this.readyState == 4 && this.status == 401) {
    //the SPOT+ rejects a PUT request without the correct authentication
    //but replies with a new nonce value
    //read the new nonce from the authentication header
    let headerString = "nonce=\"";
    let nstr = this.getResponseHeader('WWW-Authenticate');
    nstr = nstr.slice(nstr.indexOf(headerString));
    nstr = nstr.slice(headerString.length);
    nonce = nstr.substring(0, nstr.length - 1);
    nonceCounter = 1;

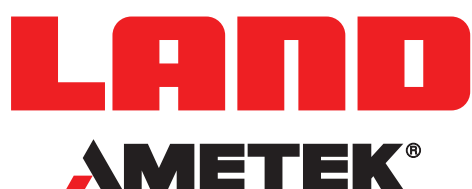
    //call the routine again
    //authentication will work next time with the new nonce value if
password is correct
    setLED();
}
    if (this.readyState == 4 && this.status == 200) {
if (this.responseText.includes("OK")) {
//the request is complete
}
}
}

//send the instruction to set the new LED state
ledhttpwrite.send(wantLED);
}

```

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Continuous product development may make it necessary to change these details without notice.

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